

MOSAIC32 ALU Control Map - Reference Sheet

14-bit micro-instruction: LUT[7:0] + ctrl[4:0] + Cin • $A_{\text{eff}} = (\text{Asel} \cdot A) \oplus \text{Ainv}$ $B_{\text{eff}} = (\text{Bsel} \cdot B) \oplus \text{Bin}$
csel=0 → Arithmetic Mode (carry chain) • csel=1 → Logic Mode (C bus)

PART 1 — Arithmetic

csel=0 → ctrl 0x00--0x0F. SUM(0x96) for sub.

#	Mnemonic	LUT	Ctrl	A_{eff}	B_{eff}	Cin	Result
1	ADD	0x96	0x05	A	B	fr	$A + B + C_{in}$
2	ADC	0x96	0x05	A	B	1	$A + B + 1$
3	INC A	0x96	0x01	A	GND	1	$A + 1$
4	INC B	0x96	0x04	GND	B	1	$B + 1$
5	DEC A	0x96	0x09	A	VCC	0	$A - 1$
6	DEC B	0x96	0x06	VCC	B	0	$B - 1$
7	NEG A	0x96	0x03	$\sim A$	GND	1	$-A$
8	NEG B	0x96	0x0C	GND	$\sim B$	1	$-B$
9	NOT A	0x96	0x03	$\sim A$	GND	0	$\sim A$
10	NOT B	0x96	0x0C	GND	$\sim B$	0	$\sim B$
11	MOV A	0x96	0x01	A	GND	0	A
12	MOV B	0x96	0x04	GND	B	0	B
13	ZERO	0x96	0x00	GND	GND	0	0
14	ONE	0x96	0x00	GND	GND	1	+1
15	ALL ONES	0x96	0x0A	VCC	VCC	1	-1
16	MINUS TWO	0x96	0x0A	VCC	VCC	0	-2
17	CMN	0x96	0x05	A	B	0	$A + B$
18	RSC	0x96	0x07	$\sim A$	B	0	$B - A - 1$
19	RSB	0x96	0x07	$\sim A$	B	1	$B - A$
20	SBB	0x96	0x0D	A	$\sim B$	0	$A - B - 1$
21	SUB	0x96	0x0D	A	$\sim B$	1	$A - B$
22	CMP	0x96	0x0D	A	$\sim B$	1	$A - B$ (flag)
23	ANOM	0x69	0x04	GND	B	0	$-B - 1$
24	ANOM	0x69	0x04	GND	B	1	$-B$
25	ANOM	0x69	0x05	A	B	0	$-A - B - 1$
26	ANOM	0x69	0x05	A	B	1	$-A - B$
27	ANOM	0x69	0x06	VCC	B	0	B
28	ANOM	0x69	0x06	VCC	B	1	$B + 1$
29	ANOM	0x96	0x0F	$\sim A$	$\sim B$	1	$\sim (A + B)$
30	ANOM	0x96	0x0F	$\sim A$	$\sim B$	0	$-A - B - 2$

Routing States

Ctrl	A_{eff} / B_{eff}	Maps To
0x00	GND / GND	ZERO / ONE
0x01	A / GND	INC A / MOV A
0x02	VCC / GND	(unused)
0x03	$\sim A$ / GND	NEG A / NOT A
0x04	GND / B	INC B / MOV B
0x05	A / B	ADD/ADC/CMN
0x06	VCC / B	DEC B / ANOM
0x07	$\sim A$ / B	RSB / RSC
0x08	GND / VCC	(unused)
0x09	A / VCC	DEC A
0x0A	VCC / VCC	ONES / MINUS2
0x0B	$\sim A$ / VCC	(unused)
0x0C	GND / $\sim B$	NEG B / NOT B
0x0D	A / $\sim B$	SUB/SBB/CMP
0x0E	VCC / $\sim B$	(unused)
0x0F	$\sim A$ / $\sim B$	ANOM

PART 2 — Logic

csel=1 → ctrl = 0x15 for all logic ops.

#	Mnemonic	LUT	Expression
25	ZERO	0x00	0
26	ONES	0xFF	1
27	MOV A	0xAA	A
28	NOT A	0x55	$\sim A$
29	MOV B	0xCC	B
30	NOT B	0x33	$\sim B$
31	MOV C	0xF0	C
32	NOT C	0x0F	$\sim C$
33	AND A,B	0x88	$A \wedge B$
34	OR A,B	0xEE	$A \vee B$
35	NAND A,B	0x77	$\sim (A \wedge B)$
36	NOR A,B	0x11	$\sim (A \vee B)$
37	XOR A,B	0x66	$A \oplus B$
38	XNOR A,B	0x99	$\sim (A \oplus B)$
39	AND A,C	0xA0	$A \wedge C$
40	OR A,C	0xFA	$A \vee C$
41	NAND A,C	0x5F	$\sim (A \wedge C)$
42	NOR A,C	0x05	$\sim (A \vee C)$
43	XOR A,C	0x5A	$A \oplus C$
44	XNOR A,C	0xA5	$\sim (A \oplus C)$
45	AND B,C	0xC0	$B \wedge C$
46	OR B,C	0xFC	$B \vee C$
47	NAND B,C	0x3F	$\sim (B \wedge C)$
48	NOR B,C	0x03	$\sim (B \vee C)$
49	XOR B,C	0x3C	$B \oplus C$
50	XNOR B,C	0xC3	$\sim (B \oplus C)$
51	BIC A,B	0x22	$A \wedge \sim B$
52	BIC B,A	0x44	$\sim A \wedge B$
53	ORN A,B	0xBB	$A \vee \sim B$
54	IMP A→B	0xDD	$\sim A \vee B$
55	BIC A,C	0x0A	$A \wedge \sim C$
56	BIC C,A	0x50	$\sim A \wedge C$

PART 2 — Continued

#	Mnemonic	LUT	Expression
57	ORN A,C	0xAF	$A \vee \sim C$
58	IMP A→C	0xF5	$\sim A \vee C$
59	BIC B,C	0x0C	$B \wedge \sim C$
60	BIC C,B	0x30	$\sim B \wedge C$
61	ORN B,C	0xCF	$B \vee \sim C$
62	IMP B→C	0xF3	$\sim B \vee C$
63	AND3	0x80	$A \wedge B \wedge C$
64	OR3	0xFE	$A \vee B \vee C$
65	NAND3	0x7F	$\sim (A \wedge B \wedge C)$
66	NOR3	0x01	$\sim (A \vee B \vee C)$
67	XOR3 / SUM	0x96	$A \oplus B \oplus C$
68	XNOR3	0x69	$\sim (A \oplus B \oplus C)$
69	MAJ	0xE8	$AB \vee BC \vee AC$
70	MIN	0x17	$\sim (AB \vee BC \vee AC)$
71	MUX C?B:A	0xCA	$C ? B : A$
72	MUX B?C:A	0xE2	$B ? C : A$
73	MUX A?C:B	0xE4	$A ? C : B$
74	AO21	0xEA	$A \vee (B \wedge C)$
75	AO21	0xEC	$B \vee (A \wedge C)$
76	AO21	0xF8	$C \vee (A \wedge B)$
77	OA21	0xA8	$A \wedge (B \vee C)$
78	OA21	0xC8	$B \wedge (A \vee C)$
79	OA21	0xE0	$C \wedge (A \vee B)$
80	AOI21	0x15	$\sim (A \vee (B \wedge C))$
81	AOI21	0x13	$\sim (B \vee (A \wedge C))$
82	AOI21	0x07	$\sim (C \vee (A \wedge B))$
83	OAI21	0x57	$\sim (A \wedge (B \vee C))$
84	OAI21	0x37	$\sim (B \wedge (A \vee C))$
85	OAI21	0x1F	$\sim (C \wedge (A \vee B))$